

RAUNAK DEY



IIT Madras
B.S. Degree

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SUMMARY

Data Science and Machine Learning student with a strong track record in quantitative research (WorldQuant) and competitive machine learning (Kaggle). Proficient in building robust predictive models using gradient boosting and ensemble methods for high-dimensional data. Additionally skilled in full-stack engineering (Next.js & TypeScript), designing scalable web applications with automated data pipelines and caching layers to deliver real-time analytical insights.

EDUCATION

Indian Institute of Technology
BSc Degree Level - Data Science and Applications

Chennai, Tamil Nadu
September 2022 – August 2027

SKILLS

Languages: Python, Java, TypeScript, JavaScript

Frameworks: Next.js, Flask, VueJS, Node.js, Tailwind

Machine Learning: XGBoost, LightGBM, CatBoost, Optuna

Data Science: Pandas, NumPy, Matplotlib, Sklearn

Databases & DevOps: PostgreSQL, Redis, Docker, Git

Other: Algorithmic Optimization, Financial Modeling

EXPERIENCE

Quantitative Research Consultant

April 2024 – March 2025

WorldQuant

Remote

- Analyzed and developed alphas with June 2025 performance stats: avg. 5 operators per alpha, 25 unique operators used, 2.47 fields per alpha (28 total), and 15 signals developed
- Successfully completed 4 pyramids across News, Options, Price-Volume, and Social Media datasets
- Achieved combined alpha performance of 0.82 and combined Power Pool alpha performance of 1.12, contributing to robust signal generation

PROJECTS

Finalysis 2.0 | Next.js 16, TypeScript, Redis, Tailwind CSS

May 2025 – June 2026

- Engineered a scalable full-stack stock research platform analyzing 2,300+ NSE stocks, delivering real-time financial metrics, fundamental scoring, and news sentiment for retail investors
- Designed a resilient split-backend architecture integrating Yahoo Finance APIs with Redis caching, circuit breakers, and automated CRON jobs for daily end-of-day market snapshots
- Built robust data pipelines to scrape continuous fundamental data and aggregate Google News RSS feeds, serving low-latency payloads via Next.js serverless API routes

CV-Embed | React 19, TypeScript, Vite, @react-pdf

February 2026 – July 2026

- Developed a fast, ATS-friendly resume builder using React 19 and TypeScript, featuring real-time visual previews and browser storage persistence, deployed at cv-embed.vercel.app
- Engineered reliable document export pipelines to generate downloadable PDF, DOCX, and normalized JSON files, integrated with Zod for live validation and resume quality scoring
- Built a custom Embed Toolkit and SDK v2 bridge using a postMessage event API, enabling seamless, resize-aware iframe integrations for external portfolios and placement portals

ACCOMPLISHMENTS

Kaggle Competitions

August 2025 – Till Date

Active Competitor (7 Completed Competitions)

- BirdCLEF+ 2026: Ranked 795 out of 4,094 teams by building a robust ensemble pipeline utilizing ProtoSSM and BirdNET models
- NVIDIA Nemotron Model Reasoning Challenge: Ranked 1,715 out of 4,182 teams in advanced reasoning benchmarks
- Consistently applied Gradient Boosting techniques (LGBM, XGBoost, CatBoost) and algorithmic hyperparameter tuning to optimize custom target metrics

International Quant Championship 2024

March 2024 – June 2024

WorldQuant

- Achieved global rank of 165 in Stage 1 and 184 in Stage 2 among thousands of participants worldwide
- Contributed to a team OS Score of 36,000+, leveraging 24+ data fields for advanced financial modeling
- Collaborated in a two-member team to design and implement scalable alpha strategies under competition pressure

EXTRA-CURRICULAR ACTIVITIES

Deputy Secretary*Diplomacia, the IITM Soft Skills Society*

March 2024 – August 2025

*SA Contribution Certificate***Marketing Team Volunteer***Student Placement Council, IITM BS Degree*

August 2024 – December 2024

SPC Impact Report